TD(λ) and Eligibility Traces

Master Degree in Control Systems Engineering 2024-2025
University of Padova

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Where are we?…

- Model - free prediction
  - Monte-Carlo (MC) methods $V(S_t) = V(S_t) + \alpha(G_t - V(S_t))$
  - Temporal – difference (TD) learning

$$V(S_t) = V(S_t) + \alpha(R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

- Model - free control
  - Monte-Carlo (MC) methods
  - Temporal – difference (TD) learning
    - Sarsa, Expected Sarsa, Q-learning

evaluation of action values rather than state values

$Q(S_t, A_t)$ instead of $V(S_t)$

- Off-policy methods (Importance sampling)

- Extension to n-step returns

Model-free prediction

- **Goal :** estimate the value function v_π using experience

$$v_\pi(s) = \mathbb{E}_\pi\left[G_t | S_t = s\right]$$

- **Monte Carlo** methods update with complete returns
  - Observe : $G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-t-1} R_T$
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MC target

To introduce Temporal Difference methods recall that

$$\begin{aligned} v_\pi(s) &= \mathbb{E}_\pi\left[G_t \mid S_t = s\right] \\ &= \mathbb{E}_\pi\left[R_{t+1} + \gamma G_{t+1} \mid S_t = s\right] \\ &= \mathbb{E}_\pi\left[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s\right] \end{aligned}$$

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MC target

- **Temporal Difference methods** update with single rewards:
  - Observe : R_{t+1}
  - Estimate return : $G_t \approx R_{t+1} + \gamma v_\pi(S_{t+1}) \approx R_{t+1} + \gamma V(S_{t+1})$
  - Update : $V(S_t) \leftarrow V(S_t) + \alpha\left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\right]$

TD target

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  - Note that the TD target resembles sampling Bellman equation

    $$v_\pi(s) = \mathbb{E}_\pi \left[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s\right]$$

    but using V *instead of* v_π

TD prediction

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated
Algorithm parameter: step size $\alpha \in (0, 1]$
Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $\boxed{V(terminal) = 0}$

Loop for each episode:
  Initialize S
  Loop for each step of episode:
    $A \leftarrow$ action given by π for S
    Take action A, observe R, S'
    $\boxed{V(S) \leftarrow V(S) + \alpha\big[R + \gamma V(S') - V(S)\big]}$
    $S \leftarrow S'$
  until S is terminal

For episodic environments

Update

The difference between the TD target and old estimate is the **TD error**

$$\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$

TD vs MC : Episodic vs Continuing

- *TD can learn before knowing the final outcome*
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- *TD can learn without the final outcome*
  - TD can learn from incomplete sequences
  - MC can only learn from complete sequences
  - TD works in continuing (non-terminating) environments
  - MC only works for episodic (terminating) environments

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- MC target $G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-t-1} R_T$ is unbiased estimate of $v_\pi(S_t)$

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  - Return depends on many random actions, transitions, rewards
  - TD target depends on one random action, transition, reward

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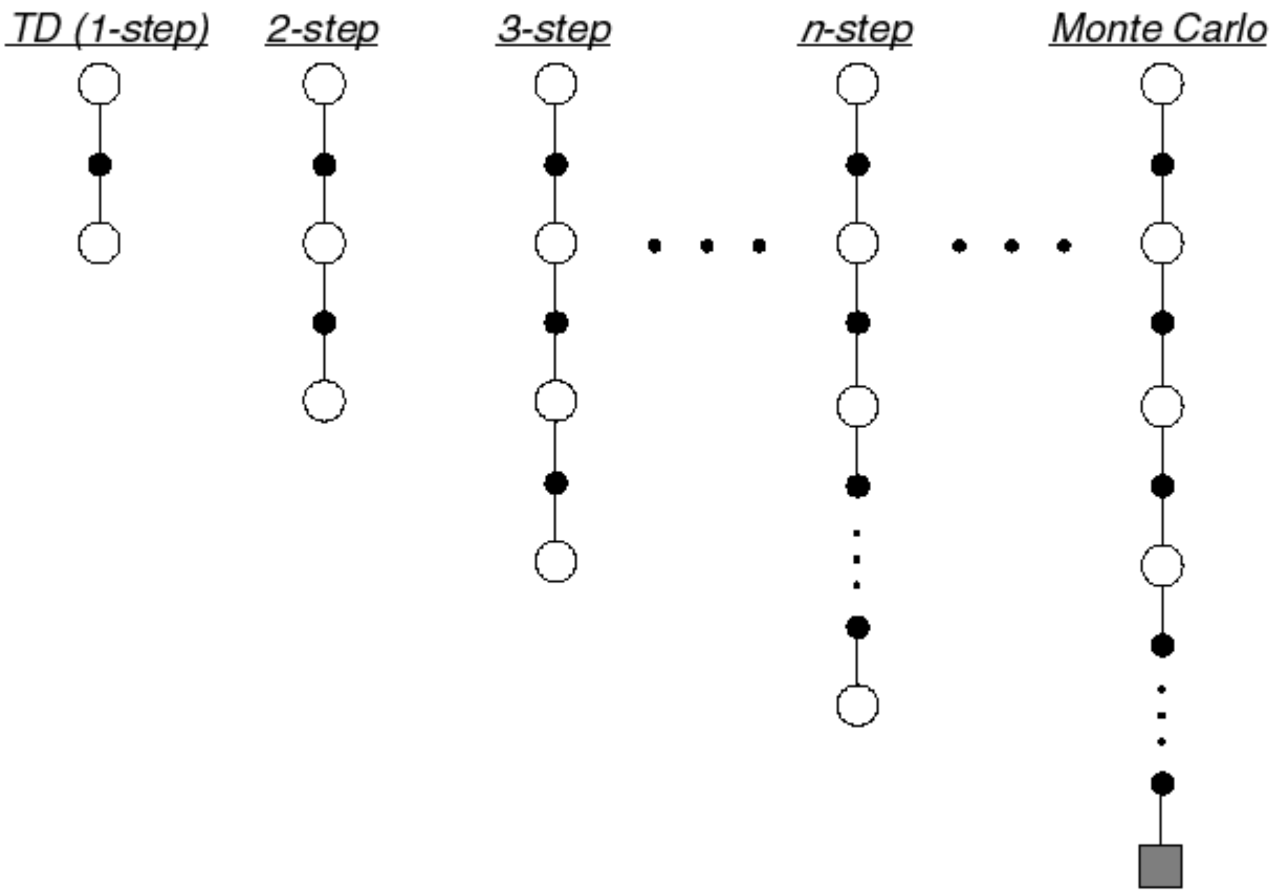
> - TD target has low variance, some bias
> - MC has high variance, zero bias

n-Step Prediction (extension of 1-step TD)

- Let TD target look *n* steps into the future



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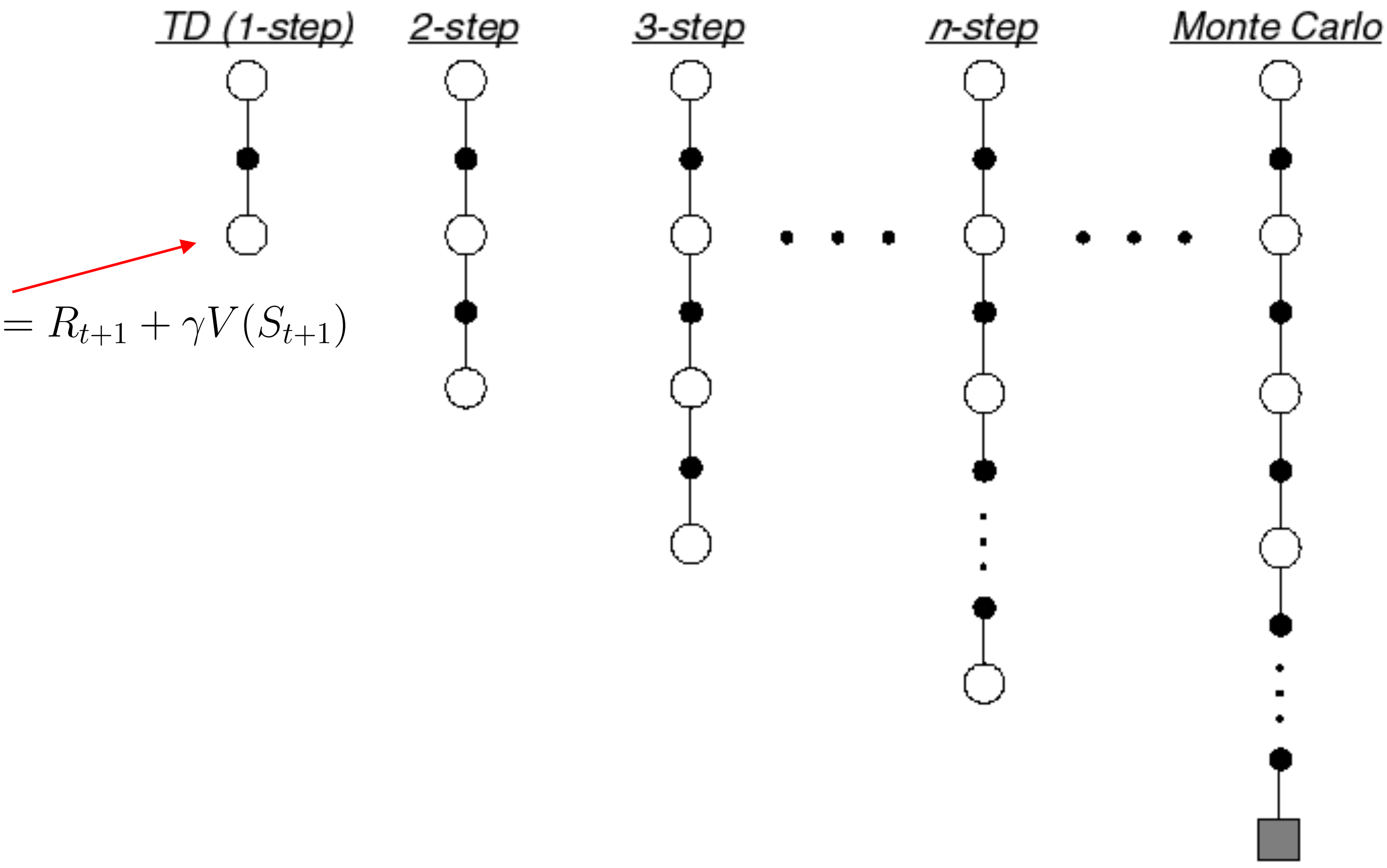

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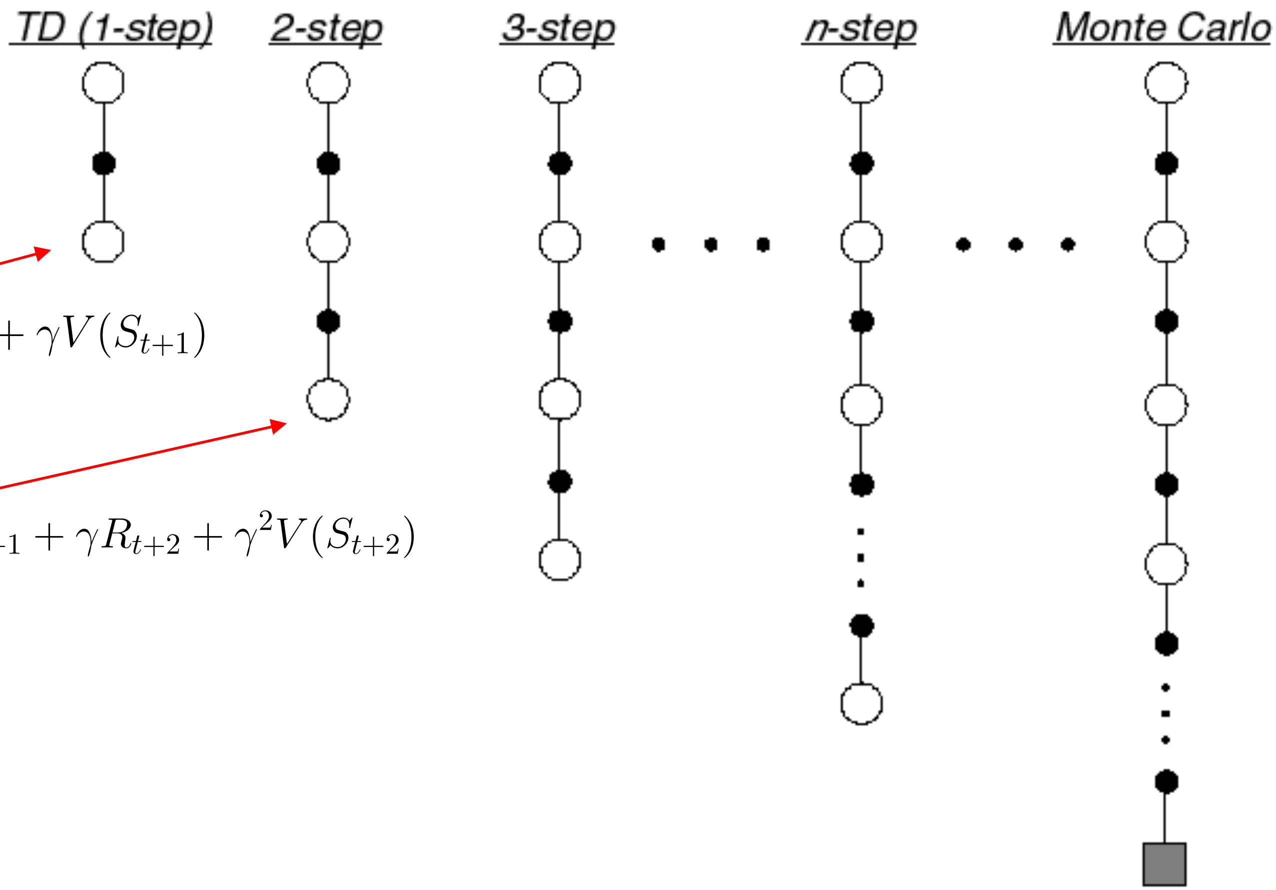

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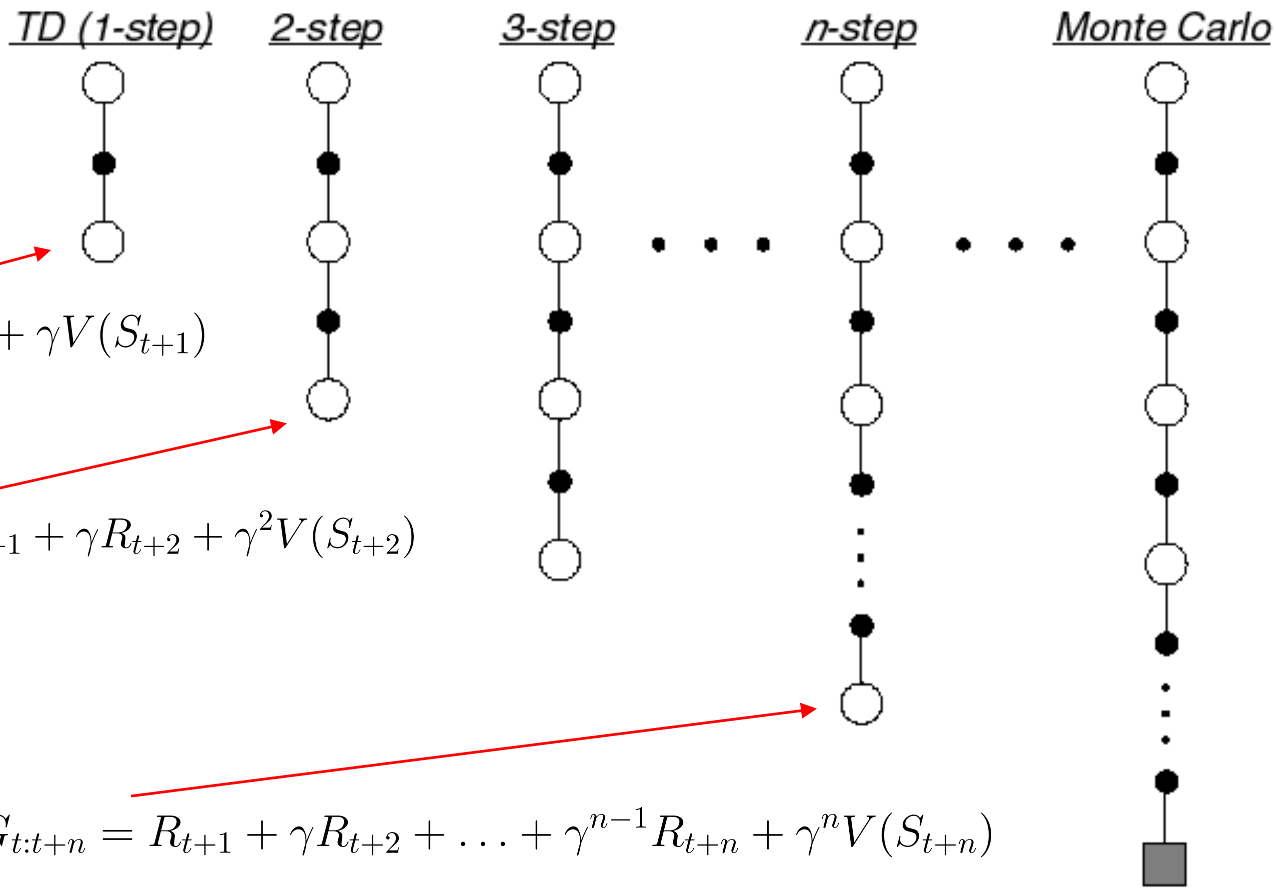

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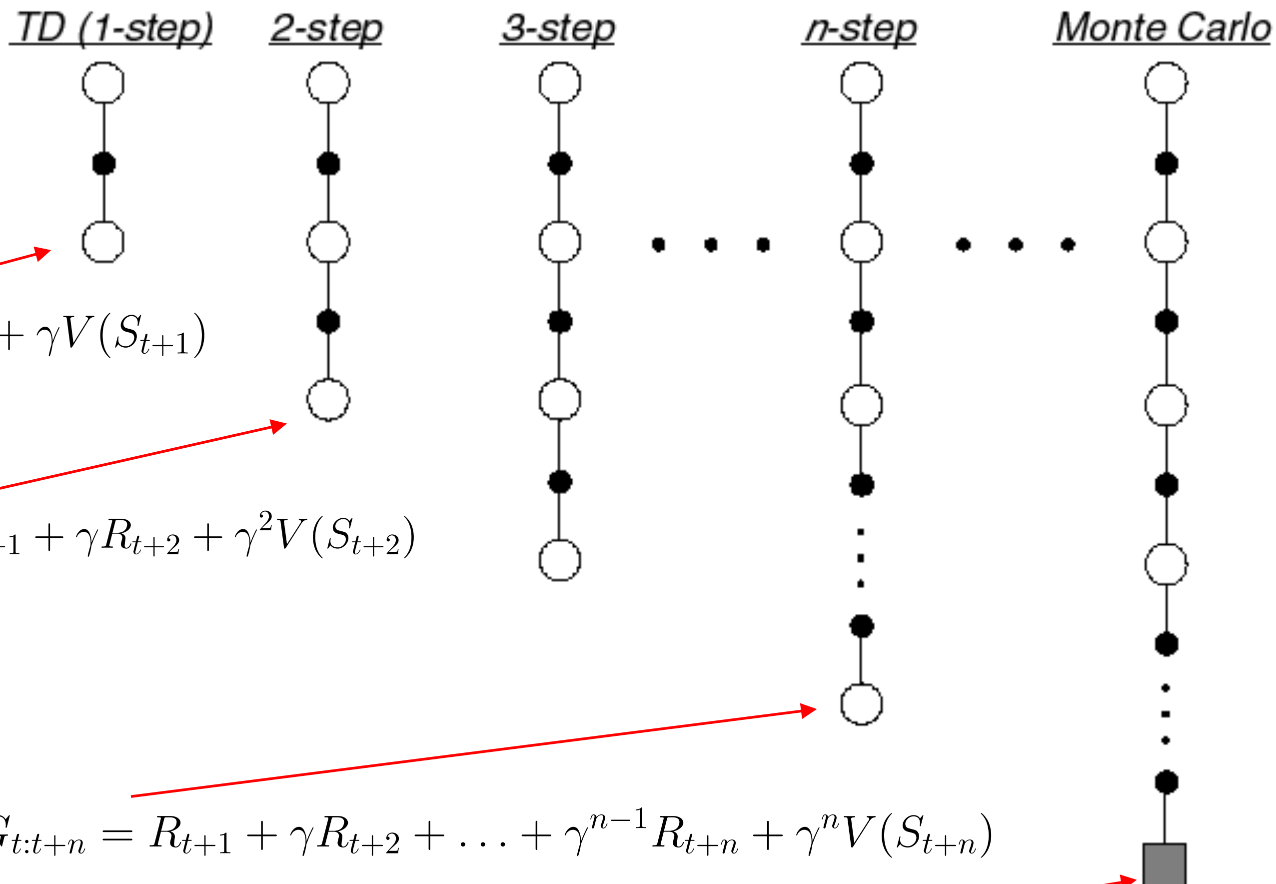

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$$G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-t-1} R_T$$

n-Step Return

- Consider the following n-step returns for $n = 1, 2, \infty$:

$$
\begin{array}{lll}
n = 1 & (TD) & G_{t:t+1} = R_{t+1} + \gamma V(S_{t+1}) \\
n = 2 & & G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2}) \\
\vdots & & \\
n = \infty & (MC) & G_t = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{T-t-1} R_T
\end{array}
$$

- Define the n-step return

$$G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \ldots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$$

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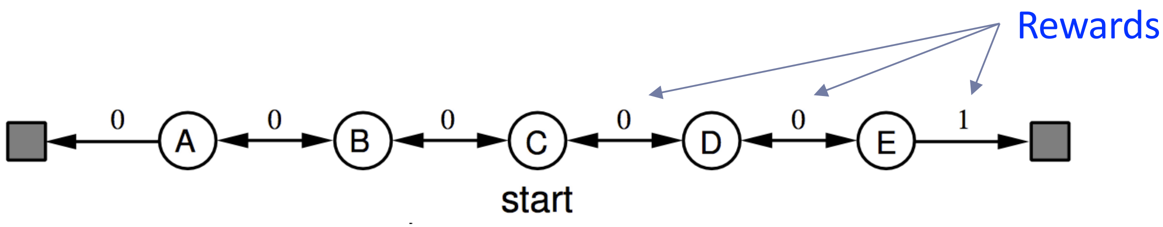
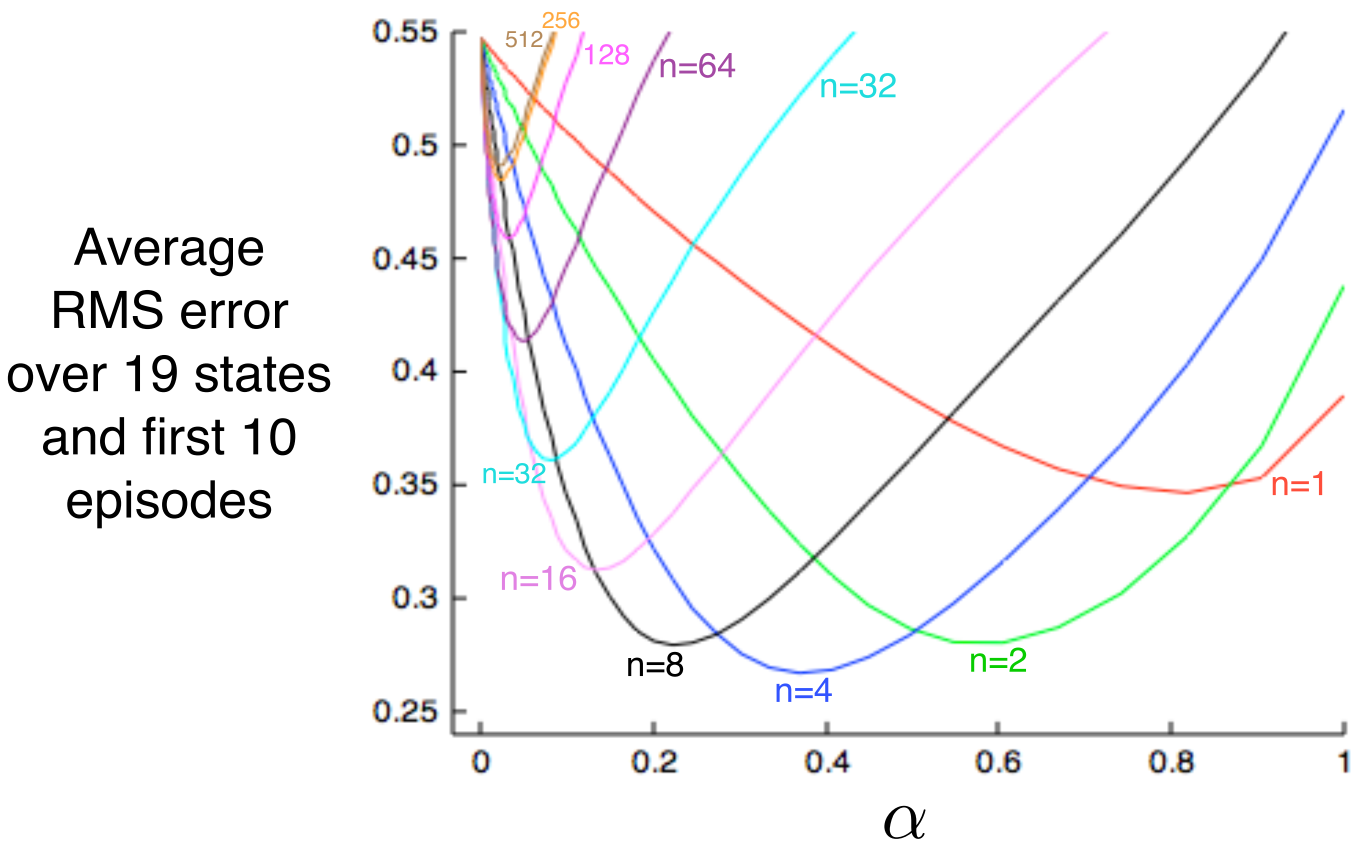
- *n*-step temporal-difference learning

$$V(S_t) \leftarrow V(S_t) + \alpha(G_{t:t+n} - V(S_t))$$

Large random walk example: which value of n is better?

The policy selects right with p=0.5 and left with p=0.5





Today's lecture…

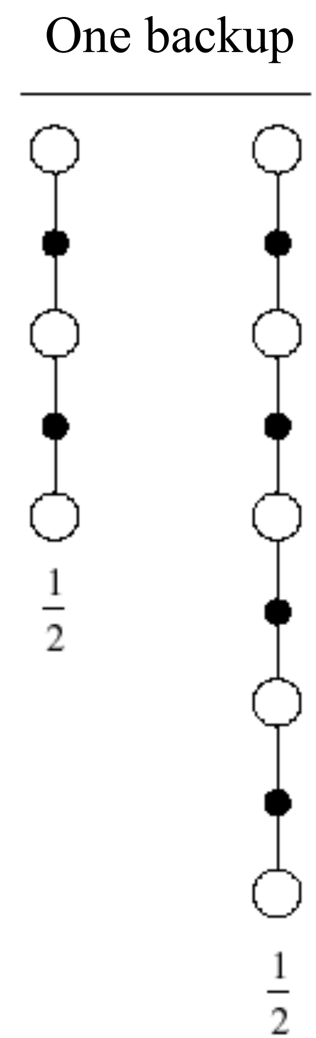
- TD(λ) algorithm for model free prediction (as generalization of n-step return)
  - Forward view
  - Backward view (Eligibility traces)
  - Equivalence between Forward and Backward views

Averaging n-Step Returns

- We can average *n*-step returns over different *n*
- e.g. average the 2-step and 4-step returns

$$\frac{1}{2}G_{t:t+2} + \frac{1}{2}G_{t:t+4}$$

- Combines information from two different time-steps



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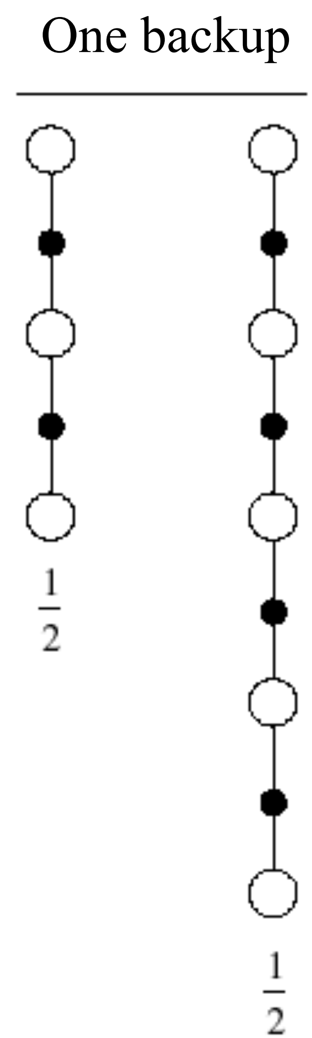
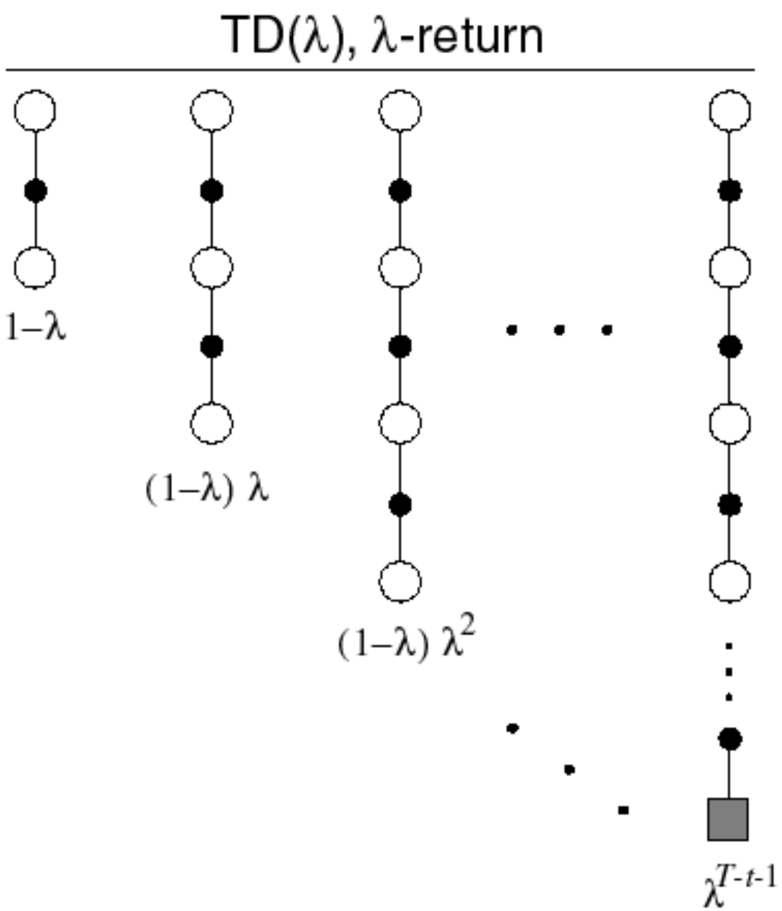
- Combines information from two different time-steps
- Can we efficiently combine information from all time-steps?



λ – return



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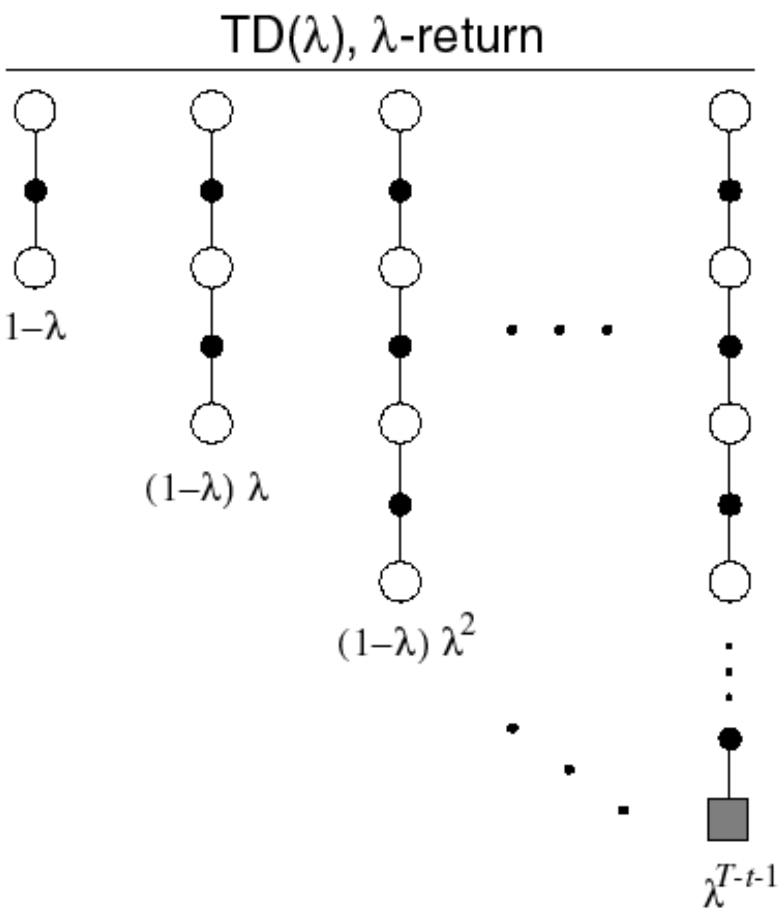
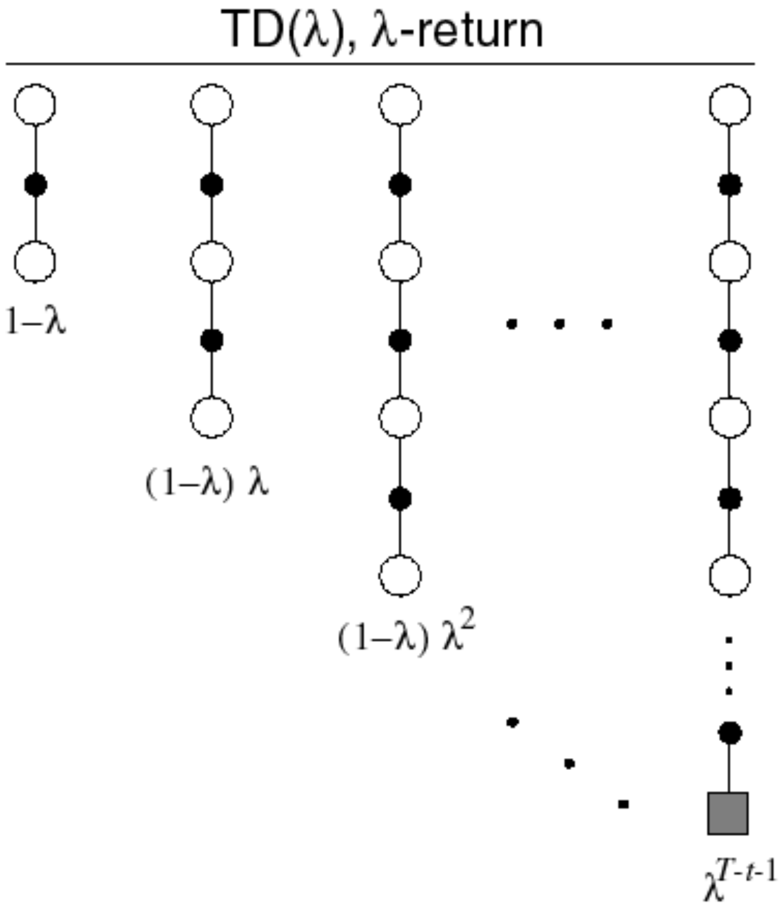


- The λ-*return* G_t^λ combines all n-step returns $G_{t:t+n}$
- Using weight $(1-\lambda)\lambda^{n-1}$

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1} G_{t:t+n}$$

λ – return



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- Forward-view TD(λ)

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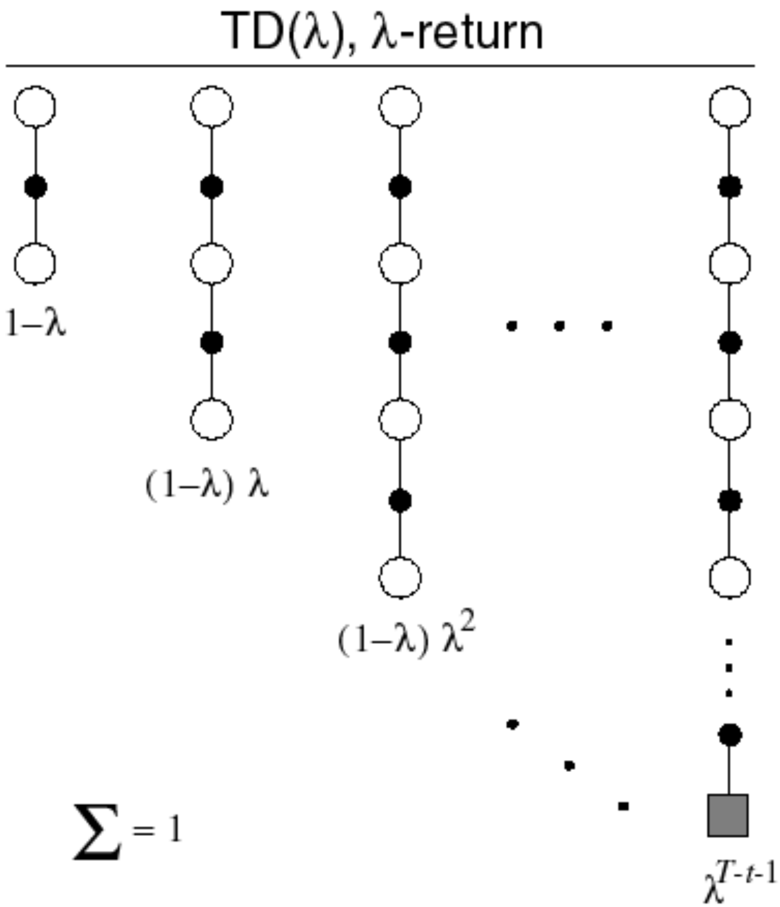
λ – return



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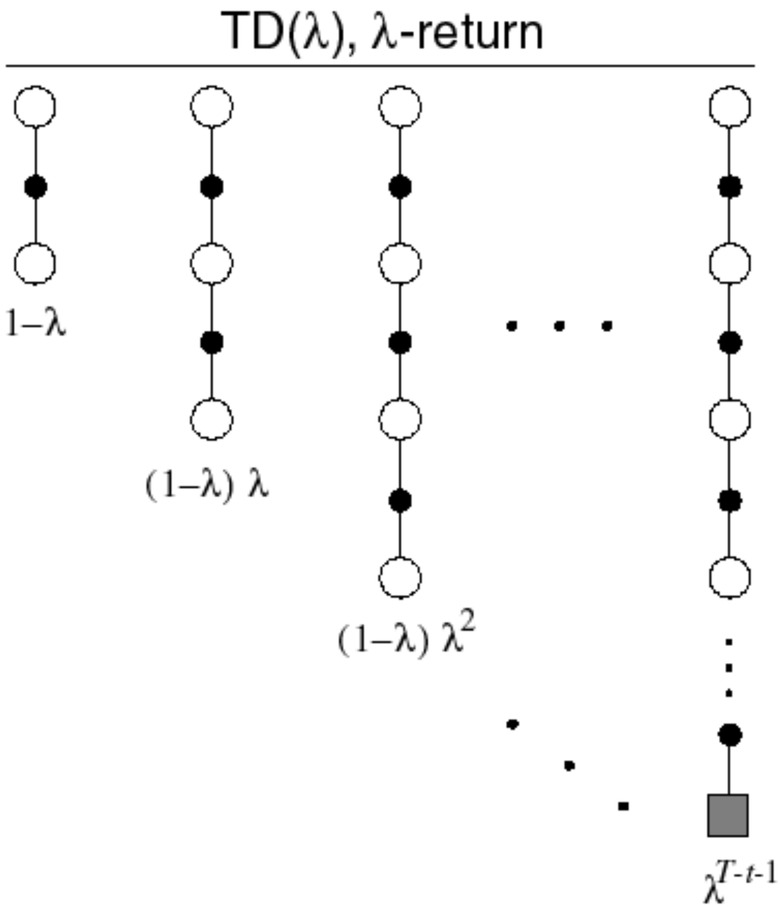
Convex combination of n-step returns (sum of weights = 1)

λ – return



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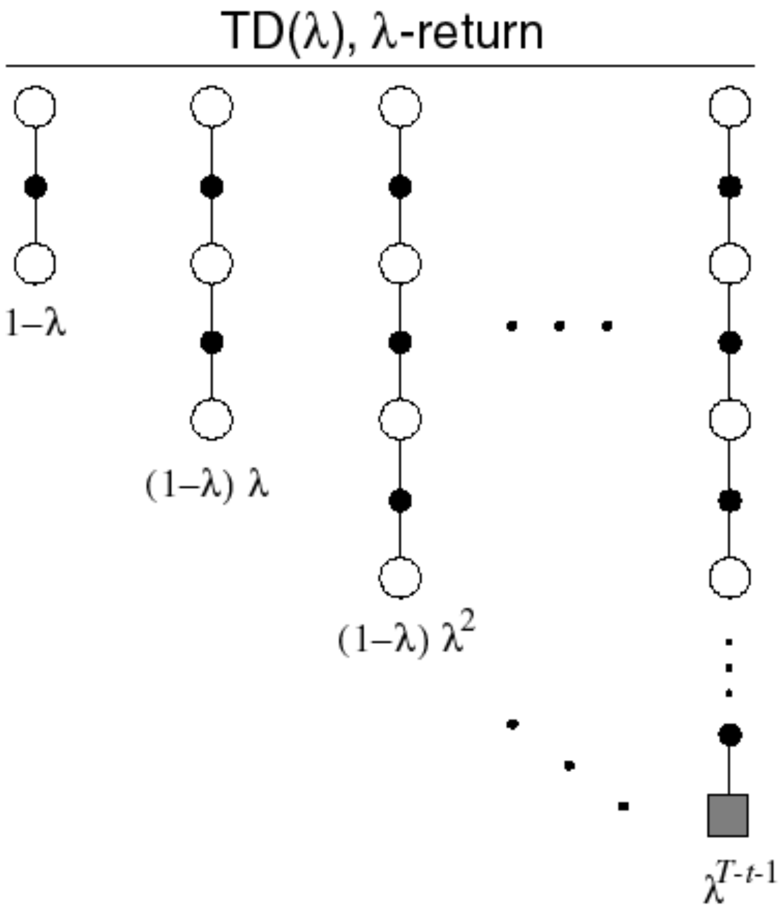
$$\sum_{n=1}^{\infty}\lambda^{n-1} = \sum_{n=0}^{\infty}\lambda^{n} = \frac{1}{1-\lambda}$$

λ – return



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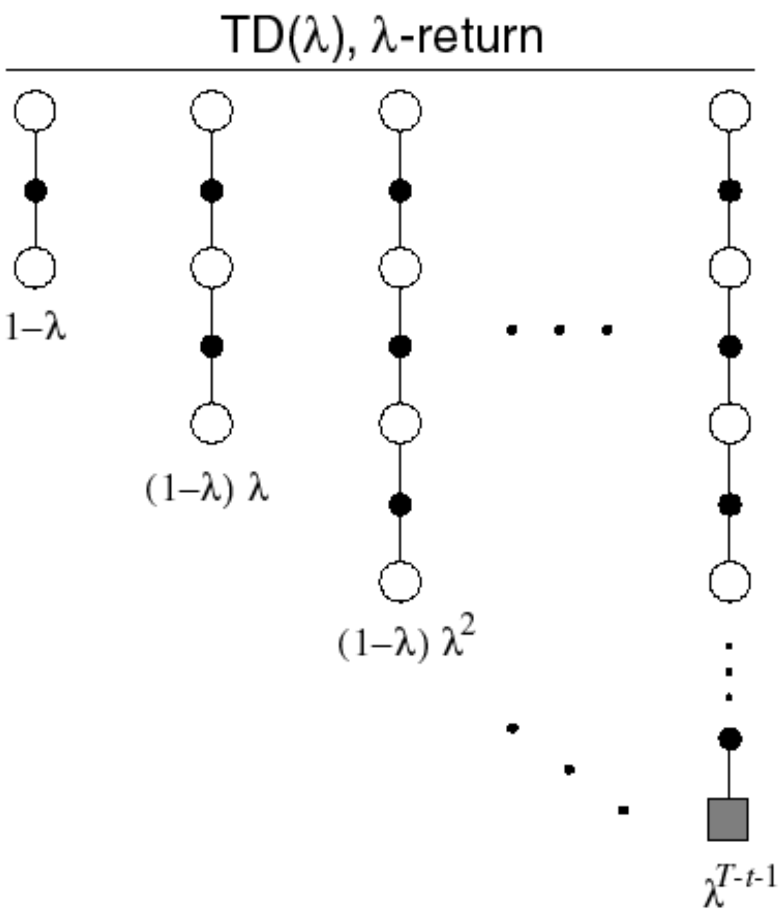
$$\sum_{n=1}^{\infty} \lambda^{n-1} = \sum_{n=0}^{\infty} \lambda^n = \frac{1}{1-\lambda} \rightarrow (1-\lambda)\sum_{n=0}^{\infty} \lambda^{n-1} = (1-\lambda)\frac{1}{1-\lambda}$$

λ – return



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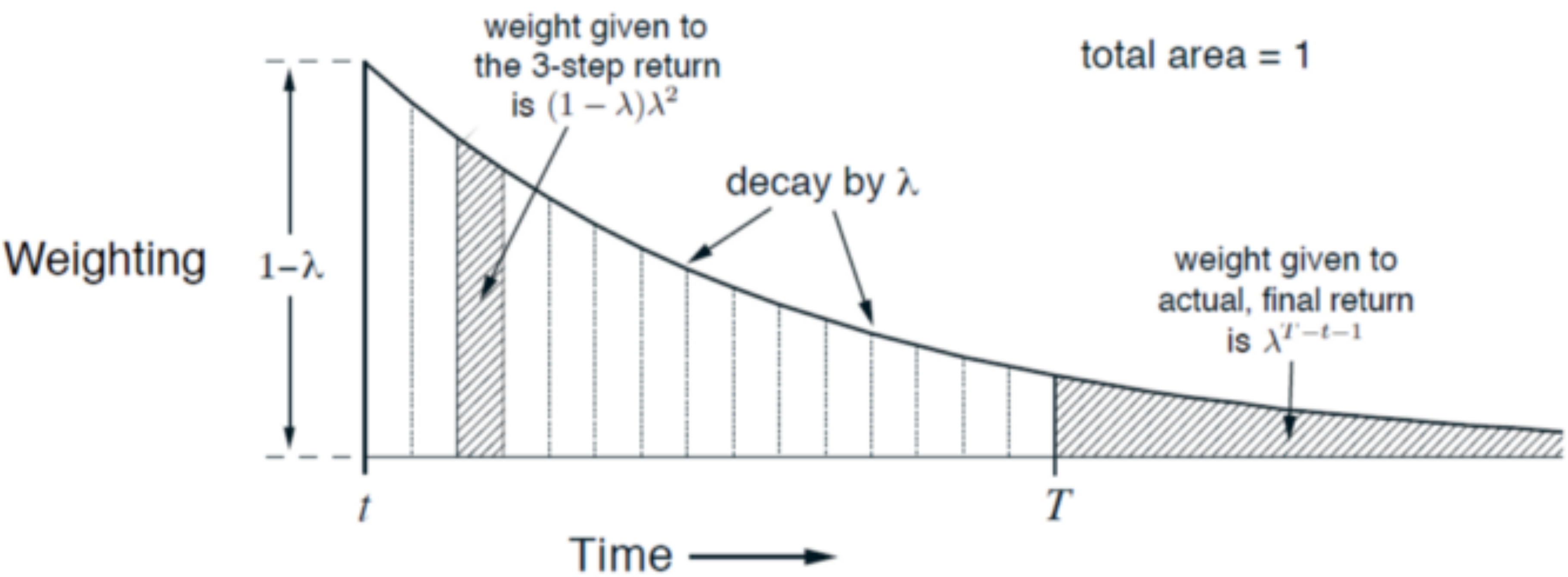
$$\sum_{n=1}^{\infty}\lambda^{n-1} = \sum_{n=0}^{\infty}\lambda^{n} = \frac{1}{1-\lambda} \Rightarrow (1-\lambda)\sum_{n=0}^{\infty}\lambda^{n-1} = (1-\lambda)\frac{1}{1-\lambda} = 1$$

TD(λ) Weighting Function



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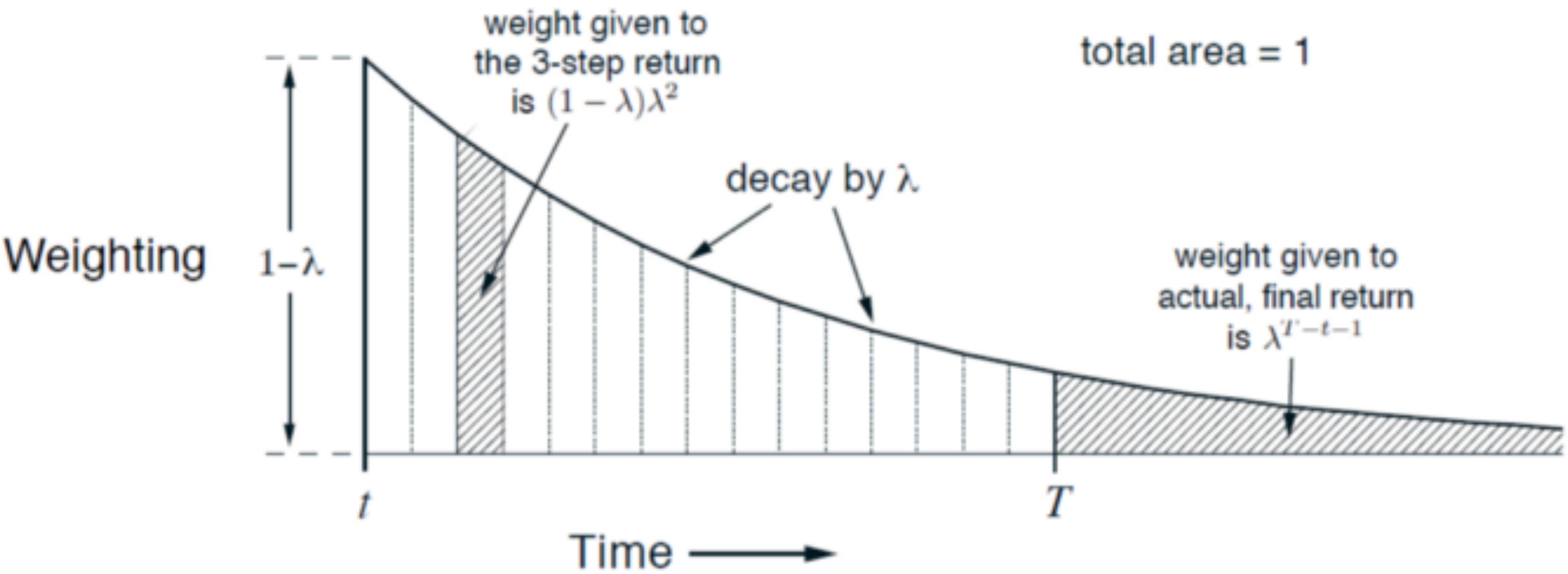

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$$G_t^{\lambda} = (1-\lambda)\sum_{n=1}^{\infty}\lambda^{n-1}\, G_{t:t+n}$$

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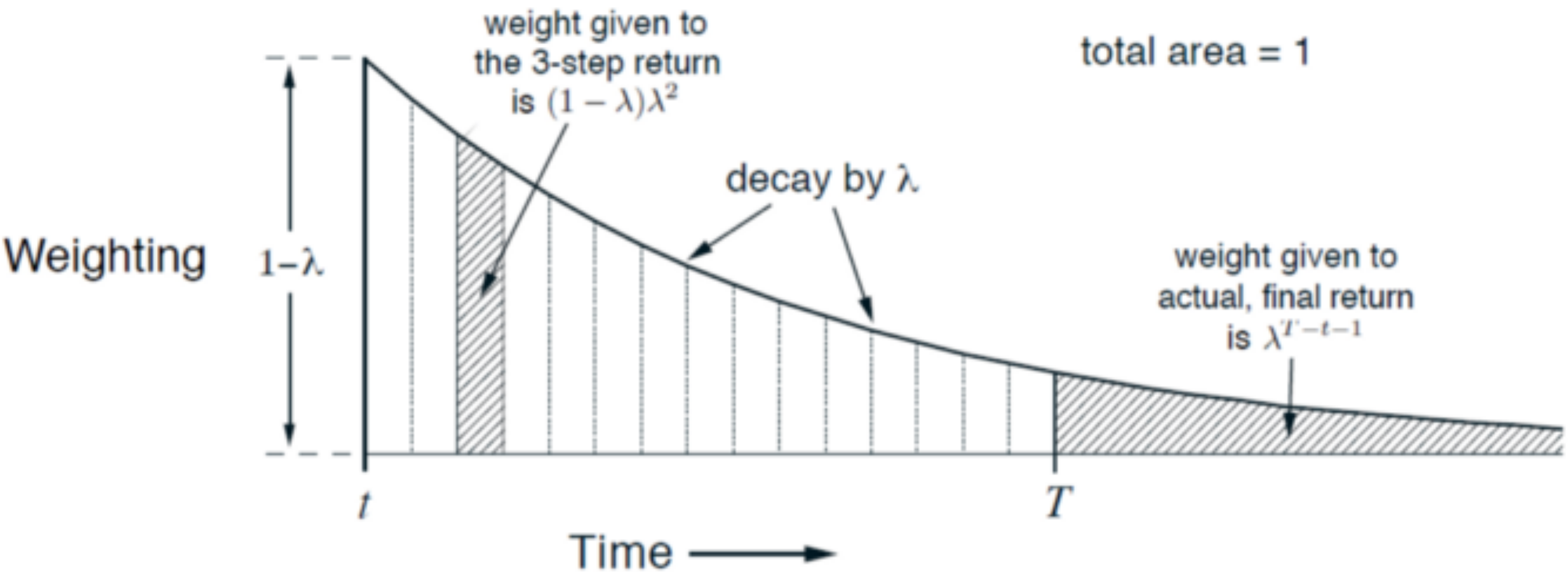
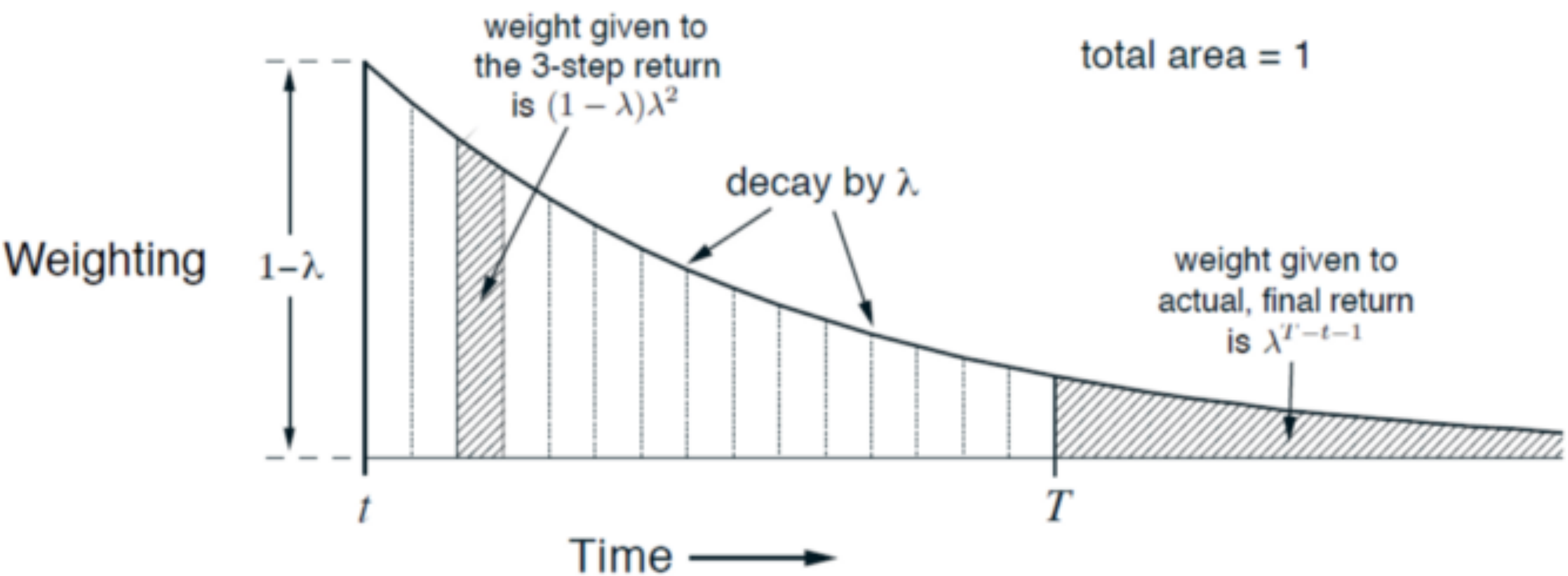


What about if an episode terminates at T?

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{T-t-1} \lambda^{n-1} G_{t:t+n} + (1-\lambda)\sum_{n=T-t}^{\infty} \lambda^{n-1} G_{t:t+n}$$

TD(λ) Weighting Function



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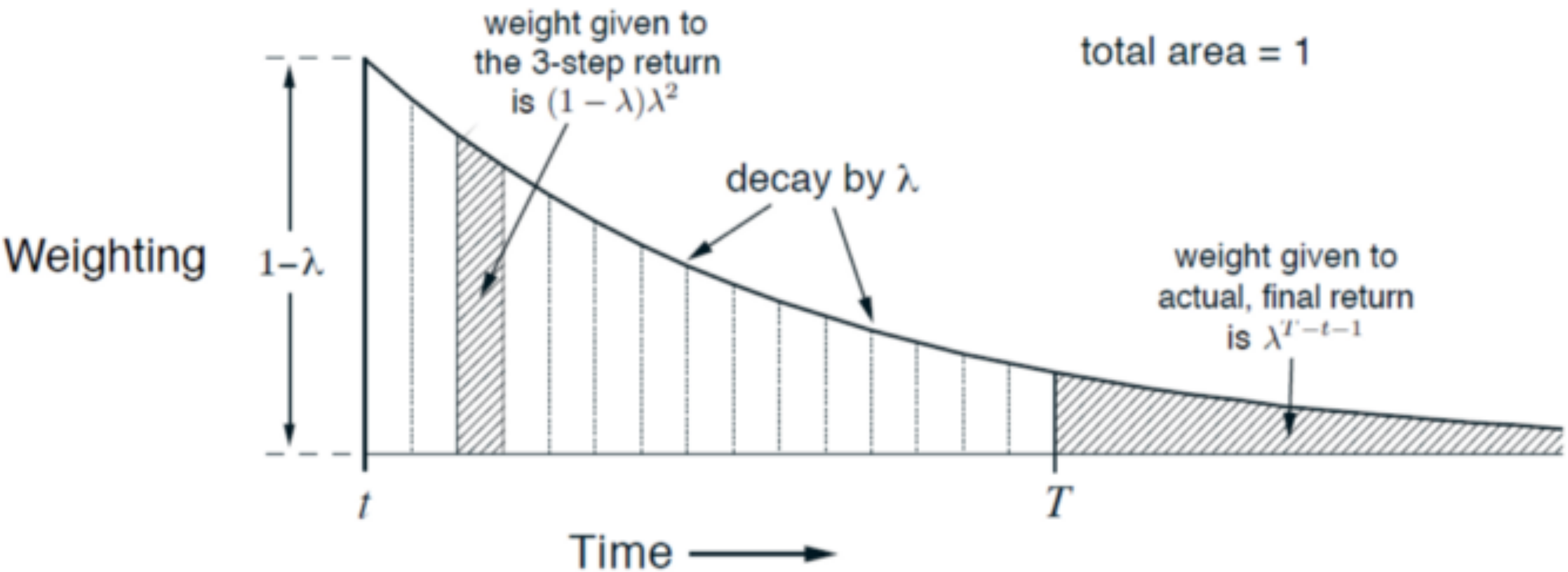
TD(λ) Weighting Function

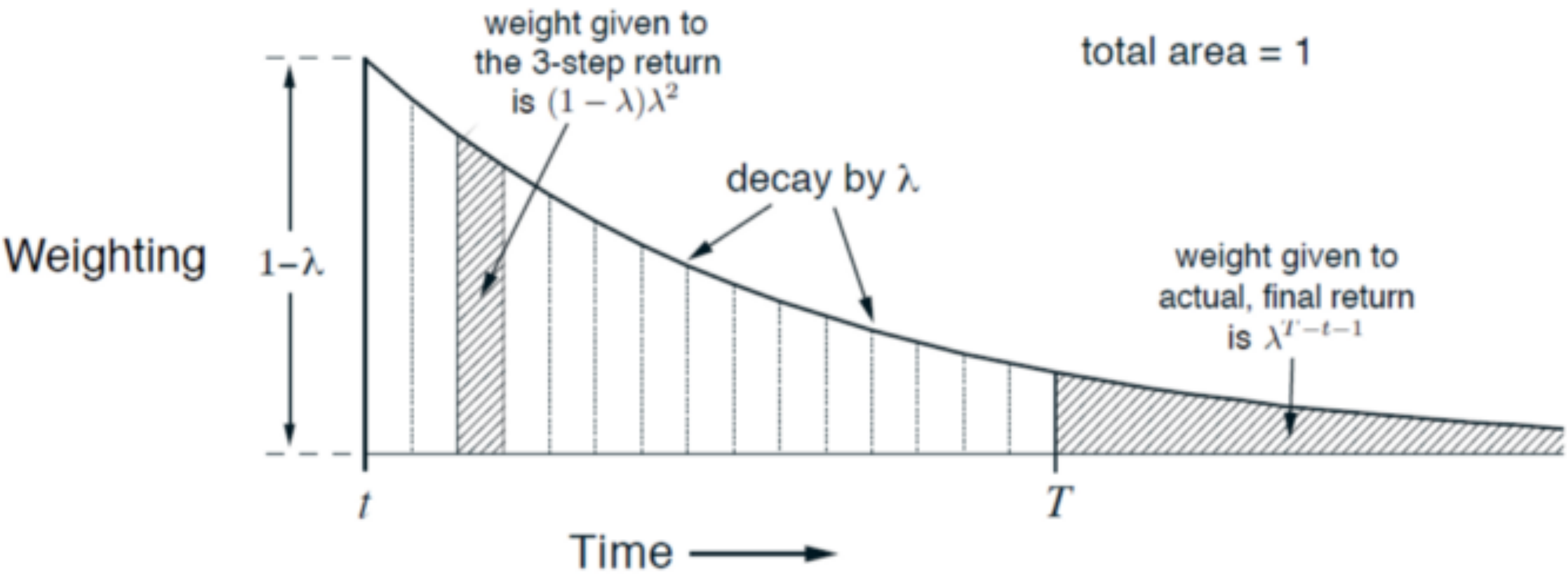


What about if an episode terminates at T?

$$G_t^\lambda = (1-\lambda)\sum_{n=1}^{T-t-1}\lambda^{n-1}G_{t:t+n} + \boxed{(1-\lambda)\sum_{n=T-t}^{\infty}\lambda^{n-1}G_t}$$

sum of geometric series starting at T-t

TD(λ) Weighting Function



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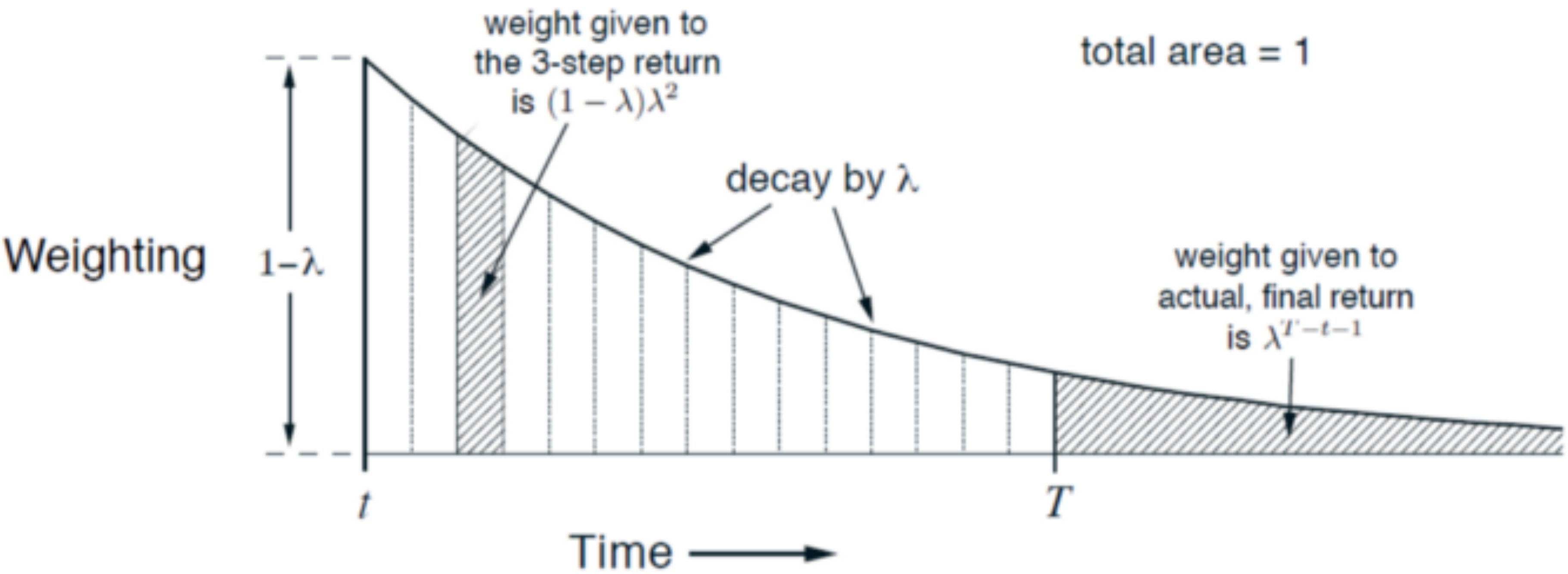
TD(λ) Weighting Function

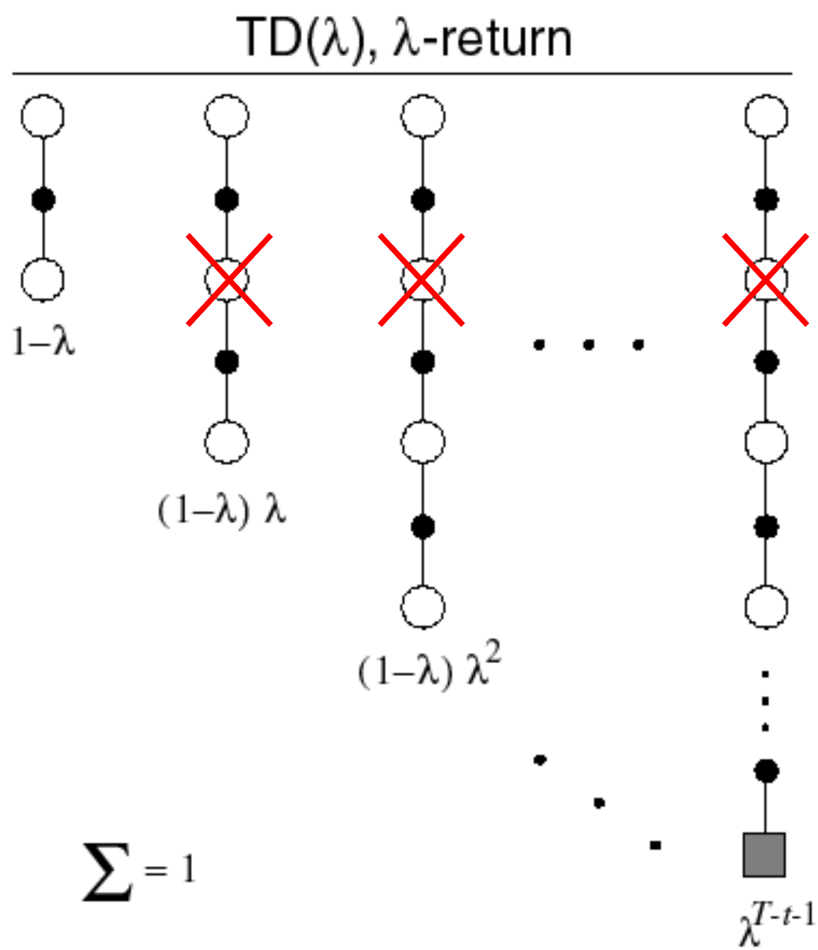


What about if an episode terminates at T?

$$G_t^{\lambda} = \underbrace{(1-\lambda)\sum_{n=1}^{T-t-1}\lambda^{n-1}G_{t:t+n}}_{\text{Until termination}} + \underbrace{\lambda^{T-t-1}G_t}_{\text{After termination}}$$

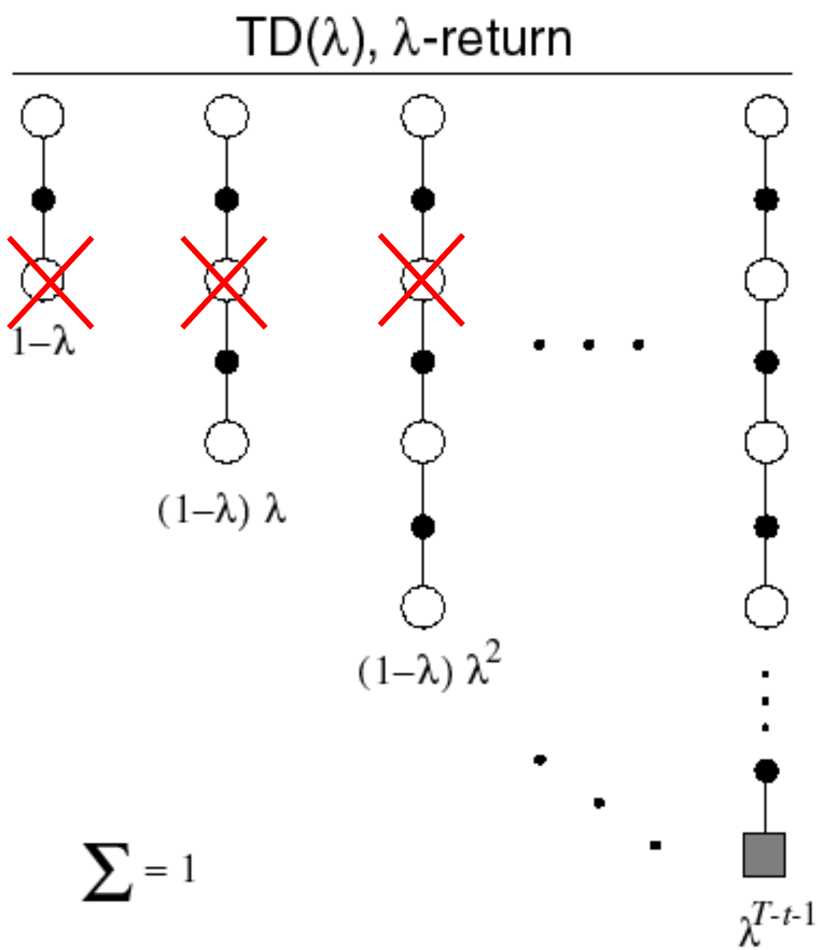
Relation to TD(0) and MC

If $\lambda = 0$ you get TD(0)

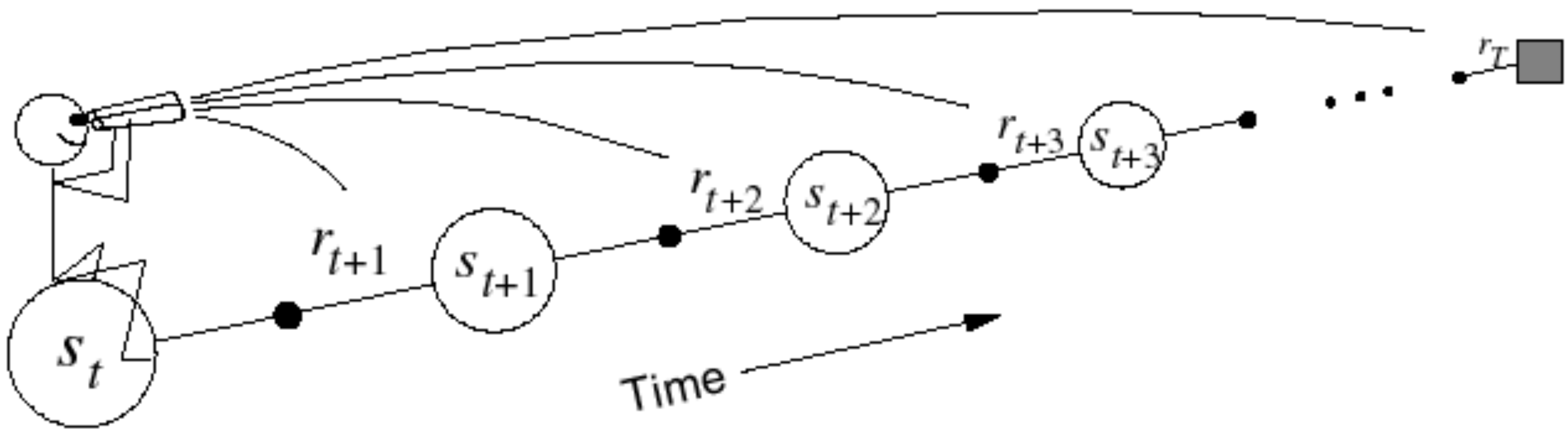


Relation to TD(0) from MC

If $\lambda = 1$ you get MC

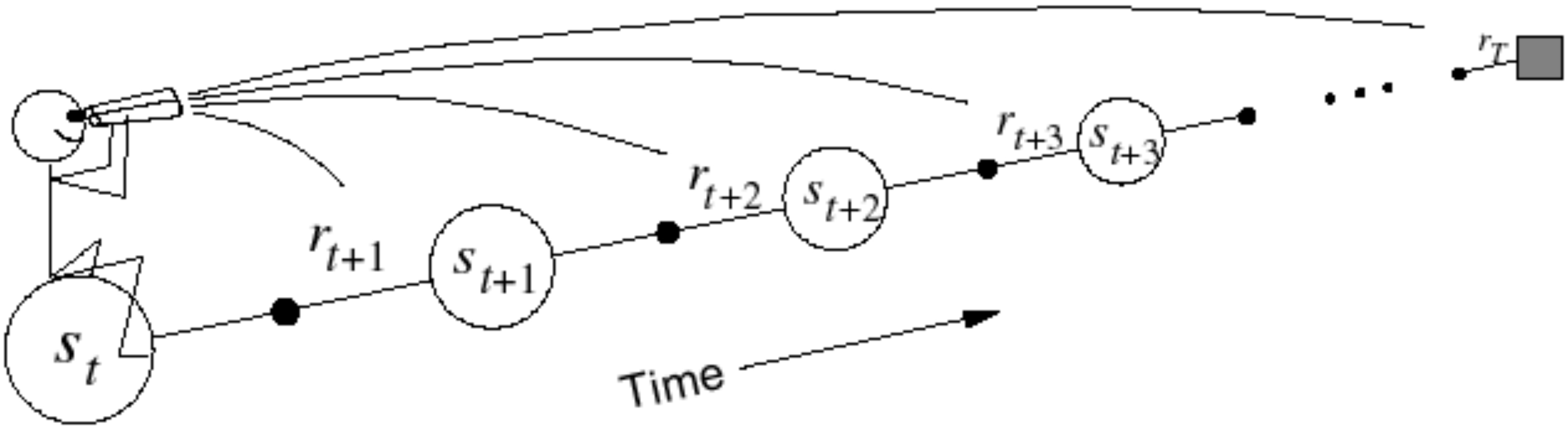


Forward - view TD(λ)



- Update value function towards the λ-return
- Forward-view looks into the future to compute G_t^λ
- Like MC, can only be computed from complete episodes

Forward - view TD(λ)



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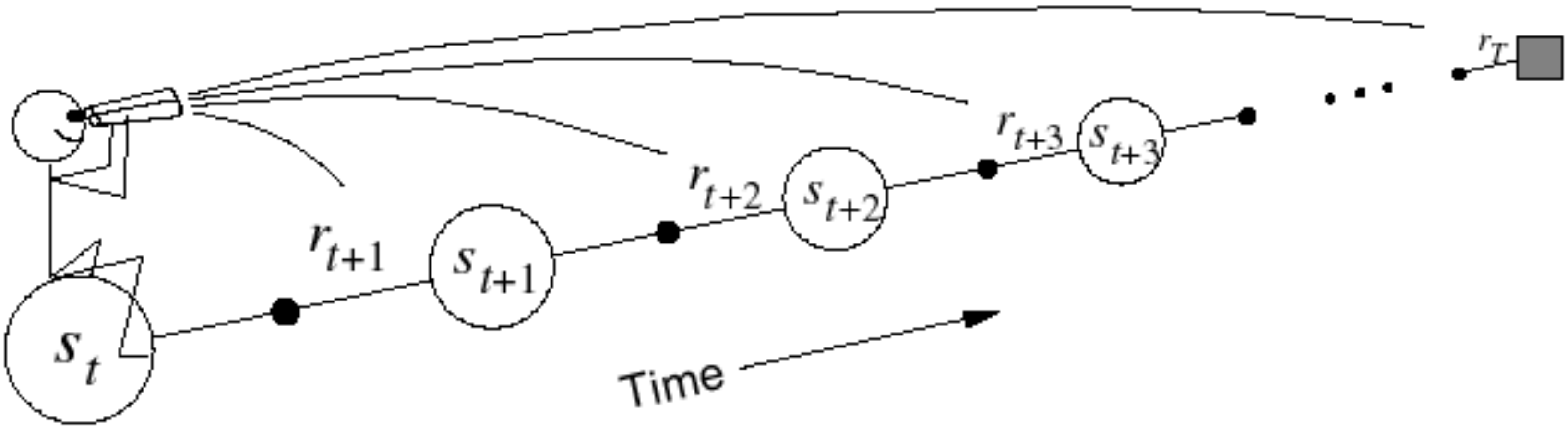
Forward views are somehow complex to implement because the update of each state depends on later events or rewards that are not available at current time

Forward - view TD(λ)



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But so far we have been doing just that! We are looking n-steps ahead…

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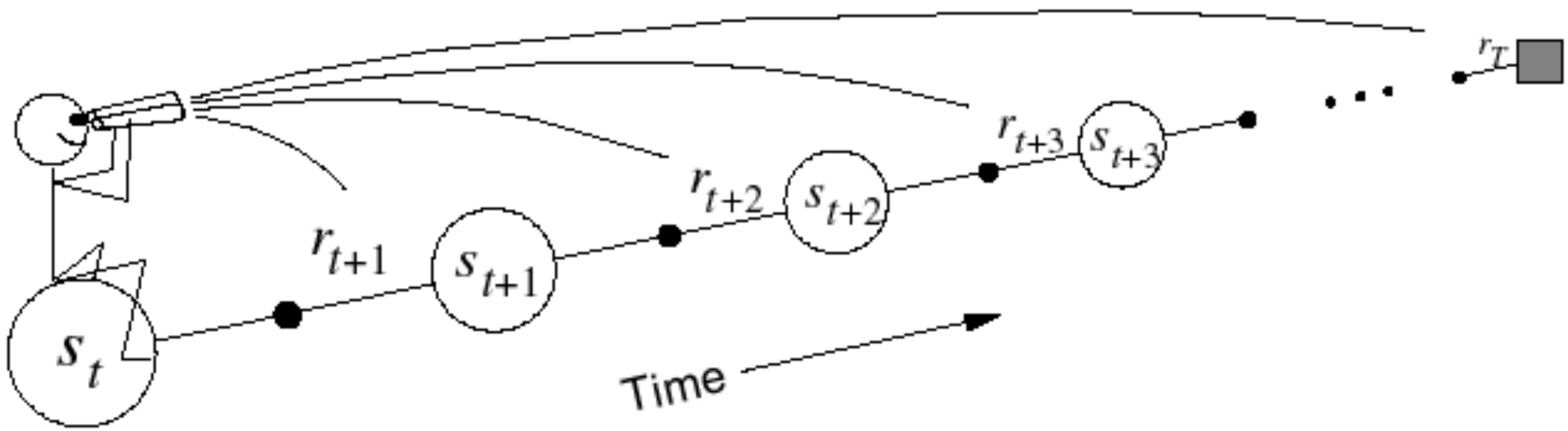

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But so far we have been doing just that! We are looking n-steps ahead…

Let us change point of view by adopting a new approach: The Backward View

Eligibility traces

Toward Backward - view TD(λ)

Some considerations on TD(λ) :

- TD(λ) methods use the λ parameter to shift from one-step TD methods to Monte Carlo methods

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- TD(λ) methods use the λ parameter to shift from one-step TD methods to Monte Carlo methods

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- Ultimately, the most compelling motivation for the λ way of mixing n-step backups is that there in a simple algorithm— **Backward-view TD(λ)** —for achieving it

Backward - view TD(λ)

- Forward view provides theory
- Backward view provides mechanism
- *Update online, every step, from incomplete sequences*
- The backward view provides a causal, incremental mechanism for approximating the forward view and, in the off-line case, for achieving it exactly.

Backward - view TD(λ) : eligibility traces

- Additional memory variable associated with each state, its **eligibility trace**
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$$E_t(s) = \gamma\lambda E_{t-1}(s), \qquad \forall s \in \mathcal{S}, s \neq S_t$$

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λ : *trace-decay parameter*

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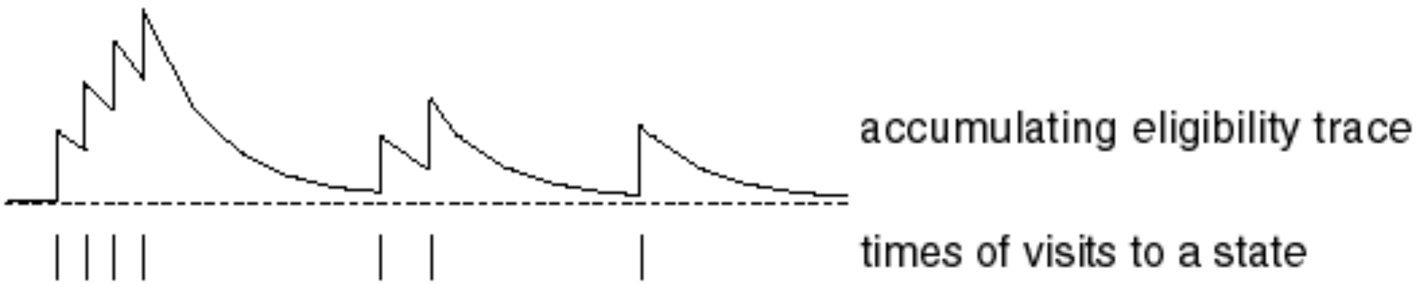
- What about the trace for S_t, the one visited at time t?

$$E_t(S_t) = \gamma\lambda E_{t-1}(S_t) + 1$$

The eligibility trace for S_t decays just like for any state, but it is incremented by 1!

Backward - view TD(λ) : eligibility traces

$$E_0(s) = 0$$
$$E_t(s) = \gamma\lambda E_{t-1}(s) + \mathbf{1}(S_t = s)$$



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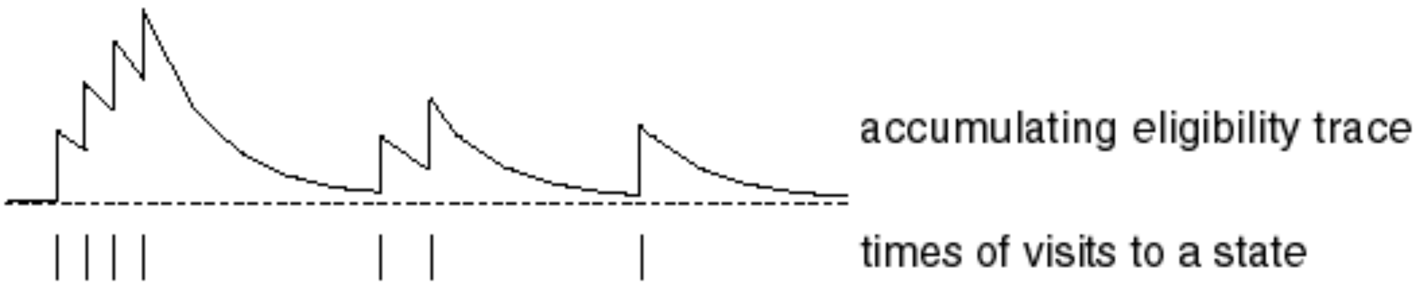

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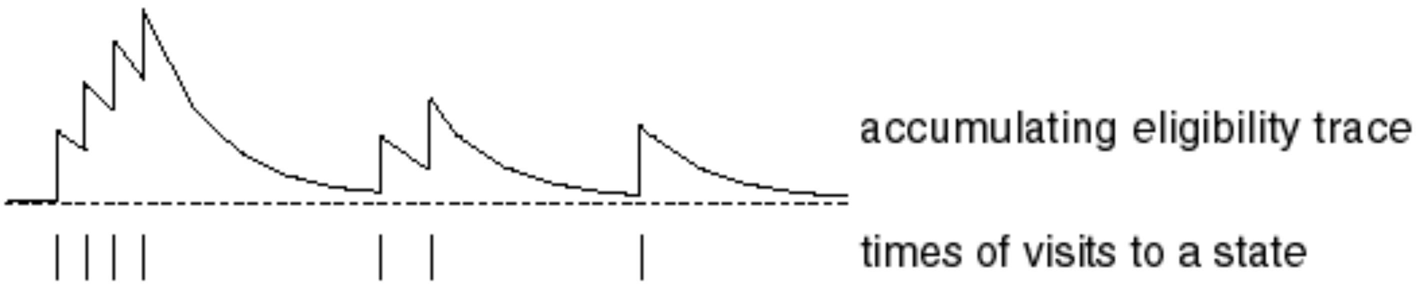

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- The traces indicate the degree to *which each state is eligible for undergoing learning changes* should a reinforcing event occur.

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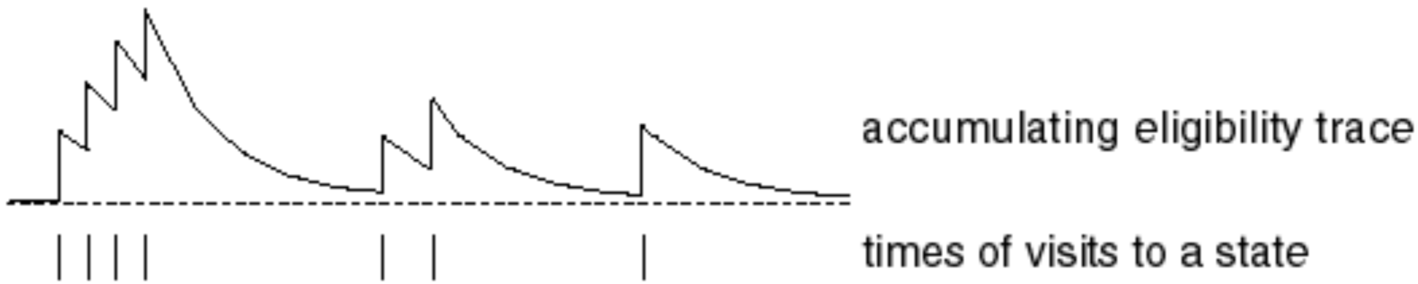


- Eligibility traces keep a simple record of which states have recently been visited, where "recently" is defined in terms of $\gamma\lambda$.

- The traces indicate the degree to which each state is eligible for undergoing learning changes should a reinforcing event occur.

- The *reinforcing events* we are concerned are one-step TD errors

$$\delta_t \;=\; R_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)$$

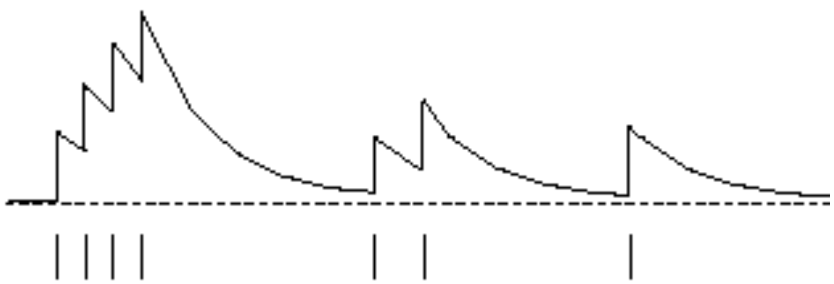
Backward - view TD(λ) : mechanism

- **TD error** : $\delta_t \;=\; R_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)$

Backward - view TD(λ) : mechanism

- **TD error** : $\delta_t \;=\; R_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)$

- **Eligibility traces update** : $E_t(s) = \gamma\lambda E_{t-1}(s) + \mathbf{1}(S_t = s)$ **for all s**



Backward - view TD(λ) : mechanism

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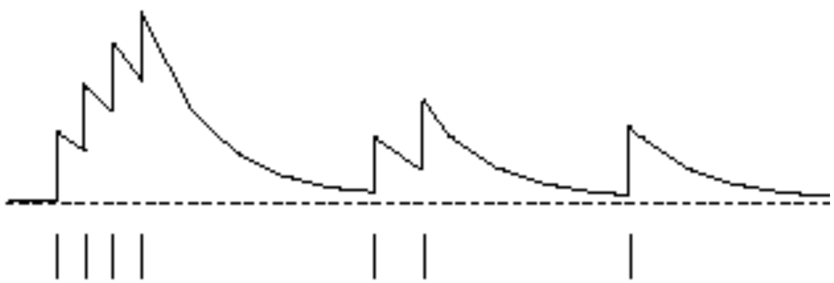
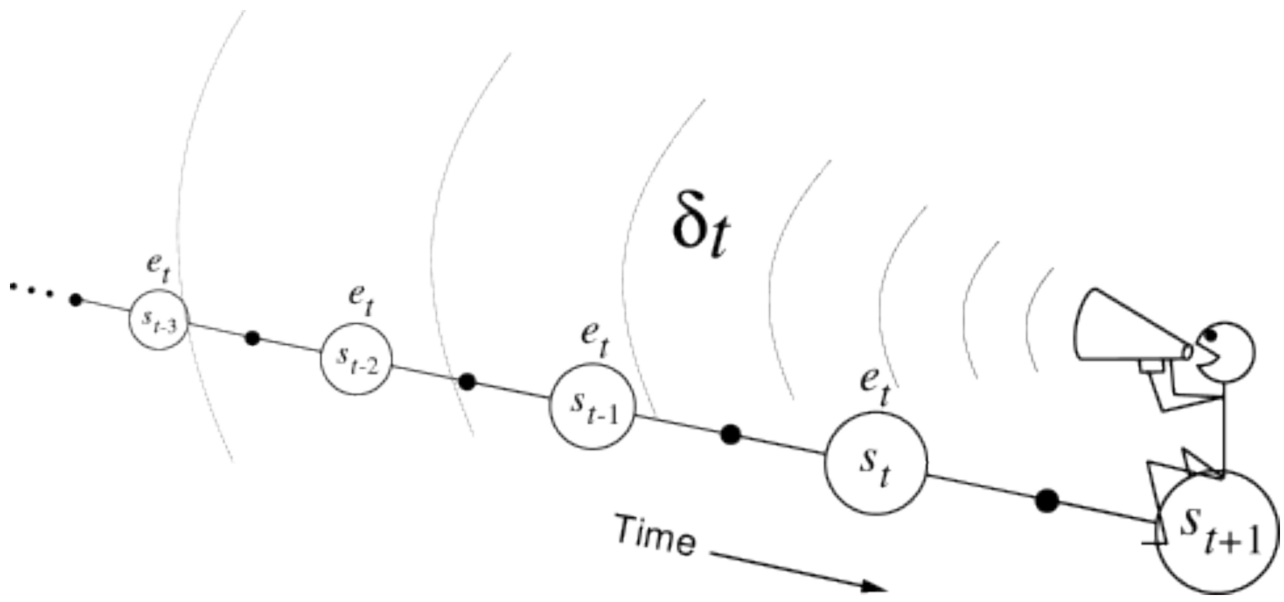


- **Value function update :** $V(s) \leftarrow V(s) + \alpha\delta_t E_t(s)$ **for all s**

Backward - view TD(λ) : mechanism

- **TD error** : $\delta_t = R_{t+1} + \gamma V_t(S_{t+1}) - V_t(S_t)$

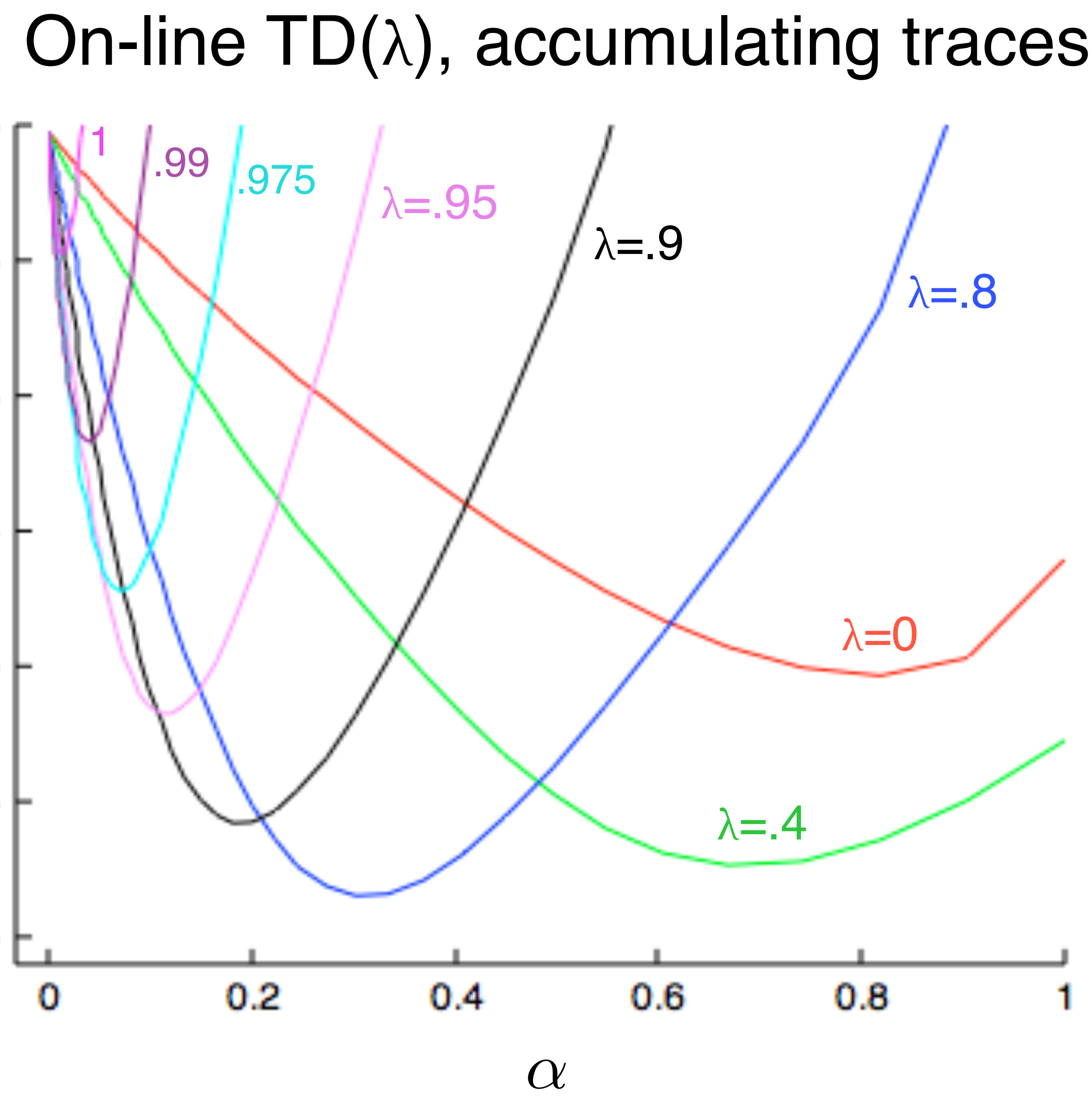
- TD error is propagated backwards but in a decaying manner - *similar to the voice that fades away with the distance (the strength of the voice decreases with the temporal distance by $\gamma\lambda$)*



Backward - view TD(λ) : algorithm

Initialize $V(s)$ arbitrarily (but set to 0 if s is terminal)
Repeat (for each episode):
  Initialize $E(s) = 0$, for all $s \in \mathcal{S}$
  Initialize S
  Repeat (for each step of episode):
    $A \leftarrow$ action given by π for S
    Take action A, observe reward, R, and next state, S'
    $\delta \leftarrow R + \gamma V(S') - V(S)$
    $E(S) \leftarrow E(S) + 1$ (accumulating traces)
    For all $s \in \mathcal{S}$:
      $V(s) \leftarrow V(s) + \alpha \delta E(s)$
      $E(s) \leftarrow \gamma \lambda E(s)$
    $S \leftarrow S'$
  until S is terminal

Backward - view TD(λ) : algorithm



Backward – view TD(λ) and TD(0)

- When $\lambda = 0$, only current state is updated

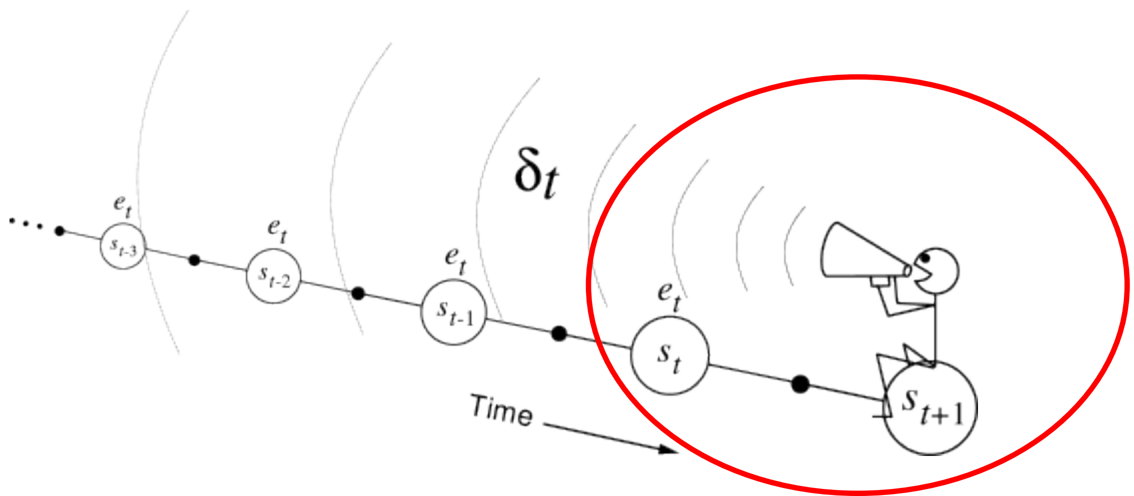
$$E_t(s) = \mathbf{1}(S_t = s)$$

$$V(s) \leftarrow V(s) + \alpha\delta_t E_t(s)$$

- This is exactly equivalent to TD(0) update

$$V(S_t) \leftarrow V(S_t) + \alpha\delta_t$$

$$\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$



Backward – view TD(λ) and MC

To be explored next lecture!

Equivalence of Forward and Backward views ?

To be explored next lecture!

Material…

- The last version of the book described *eligibility traces* in Chapter 12…
- …but it is mixed with value function approximation algorithms (in my opinion it is a bit confusing)
- A previous *in progress* version of the book (2014/1015) describes in Chapter 7 (*more and less*) the material I have presented. I will upload this *in progress* version of he book.
- *More and less* means that sometimes the notation is slightly different and that I have presented just a part of Chapter 7 of the *in progress* version 2014/2015; in particular I did not mention Dutch traces and I did not discuss Off-policy importance sampling algorithms
- The material required for the final exam is the one I have presented today in the slides (please read Chapter 7 of the *in progress* version 2014/2015 as reference for intuition and deep explanations)